

Equações:

→ Massa: $\frac{\partial \alpha_k}{\partial t} + \frac{\partial (\alpha_k u_k)}{\partial x} = 0$

→ Momento: $\frac{\partial (\alpha_k u_k)}{\partial t} + \frac{\partial (\alpha_k u_k^2)}{\partial x} = - \frac{\alpha_k}{\rho_k} \frac{\partial p_k}{\partial x} - \frac{1}{\rho_k} (p_k - p_k^i) \frac{\partial \alpha_k}{\partial x} + \alpha_k \rho_v \sin \theta$

→ Fechamento: $\alpha_l + \alpha_c = 1$

• Modelo CATHARE: $(p_k - p_k^i) = \gamma \frac{\alpha_c \alpha_l \rho_c \rho_l}{\alpha_c \rho_l + \alpha_l \rho_c} (u_c - u_l)^2$

Discretização:

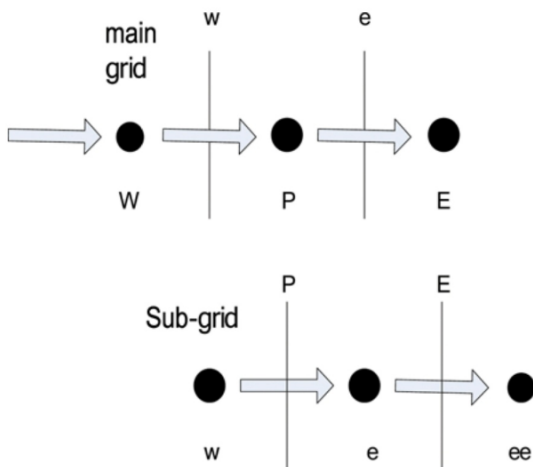


Figure 1. Staggered grid arrangement.

Massa:

$$\frac{\alpha_{k,P}^{n+1} - \alpha_{k,P}^n}{\Delta t} + \frac{\alpha_{k,e}^n u_{k,e}^n - \alpha_{k,w}^n u_{k,w}^n}{\Delta x} = 0$$

Momento:

$$\alpha_{k,e}^n \frac{(u_{k,e}^{n+1} - u_{k,e}^n)}{\Delta t} + \alpha_{k,e}^n u_{k,e}^n \frac{(u_{k,E}^{n+1} - u_{k,P}^{n+1})}{\Delta x} =$$

$$- \frac{\alpha_{k,e}^n}{\rho_{k,e}^n} \frac{(p_E^{n+1} - p_W^{n+1})}{\Delta x} + \alpha_{k,e}^n g \sin \theta + \frac{1}{\rho_{k,e}^n} (p_k - p_k^i) \frac{\alpha_{k,E}^n - \alpha_{k,P}^n}{\Delta x}$$

Correção da Pressão:

$$\frac{1}{\rho_{c,ref}} (\alpha_{c,e}^k u_{c,e}^{k+1} - \alpha_{c,w}^k u_{c,w}^{k+1}) + \frac{1}{\rho_{l,ref}} (\alpha_{l,e}^k u_{l,e}^{k+1} - \alpha_{l,w}^k u_{l,w}^{k+1}) + \frac{\Delta x}{\Delta t} \left\{ \frac{\alpha_{c,P}^k - \alpha_{c,P}^n}{\rho_{c,ref}} + \frac{\alpha_{l,P}^k - \alpha_{l,P}^n}{\rho_{l,ref}} \right\} = 0$$

→ $u_{k,e}^{k+1} = u_{k,e}^k + \frac{\Delta t}{\Delta x \cdot \rho_{k,e}^n} (sp_P - sp_E)$

Substituindo:

$$\frac{\alpha_{c,e}^k u_{c,e}^k}{\rho_{c,ref}} + \frac{\alpha_{c,e}^k \Delta t}{\rho_{c,ref} \cdot \rho_{c,e}^n \cdot \Delta x} (sp_P - sp_E) - \frac{\alpha_{c,w}^k u_{c,w}^k}{\rho_{c,ref}} - \frac{\alpha_{c,w}^k \Delta t}{\rho_{c,ref} \cdot \rho_{c,w}^n \cdot \Delta x} (sp_W - sp_P) + \frac{\alpha_{l,e}^k u_{l,e}^k}{\rho_{l,ref}}$$

$$+ \frac{\alpha_{l,e}^k \cdot \Delta t}{\rho_{l,ref} \cdot \rho_{l,e}^m \cdot \Delta x} (\delta p_p - \delta p_e) - \frac{\alpha_{l,w}^k \cdot u_{l,w}^k}{\rho_{l,ref}} - \frac{\alpha_{l,w}^k \cdot \Delta t}{\rho_{l,ref} \cdot \rho_{l,w}^m \cdot \Delta x} (\delta p_w - \delta p_p) + b = 0$$

Reorganizando para montar o sistema linear:

$$\left(\frac{\alpha_{c,e}^k \cdot \Delta t}{\rho_{c,ref} \cdot \rho_{c,e}^m \cdot \Delta x} + \frac{\alpha_{c,w}^k \cdot \Delta t}{\rho_{c,ref} \cdot \rho_{c,e}^m \cdot \Delta x} + \frac{\alpha_{l,e}^k \cdot \Delta t}{\rho_{l,ref} \cdot \rho_{l,e}^m \cdot \Delta x} + \frac{\alpha_{l,w}^k \cdot \Delta t}{\rho_{l,ref} \cdot \rho_{l,e}^m \cdot \Delta x} \right) \cdot \delta p_p + \left(\frac{-\alpha_{c,e}^k \cdot \Delta t}{\rho_{c,ref} \cdot \rho_{c,e}^m \cdot \Delta x} - \frac{\alpha_{l,e}^k \cdot \Delta t}{\rho_{l,ref} \cdot \rho_{l,e}^m \cdot \Delta x} \right) \cdot \delta p_e + \left(\frac{-\alpha_{c,w}^k \cdot \Delta t}{\rho_{c,ref} \cdot \rho_{c,w}^m \cdot \Delta x} - \frac{\alpha_{l,w}^k \cdot \Delta t}{\rho_{l,ref} \cdot \rho_{l,w}^m \cdot \Delta x} \right) \cdot \delta p_w = - \frac{\alpha_{c,e}^k \cdot u_{c,e}^k}{\rho_{c,ref}} - \frac{\alpha_{c,w}^k \cdot u_{c,w}^k}{\rho_{c,ref}} - \frac{\alpha_{l,e}^k \cdot u_{l,e}^k}{\rho_{l,ref}} - \frac{\alpha_{l,w}^k \cdot u_{l,w}^k}{\rho_{l,ref}} - b$$